MoonWalker Weekly Meeting

10/8-10/15

Elijah, Martin, Rowel

Progress Update

- Elijah
  - Reviewing Omniverse ActionGraph and BehaviorScript
  - Working on OmniverseBL
- Martin
  - Set up Unity to work, even with some missing assets
  - Set up unity_hsml to work with unitybl
- Rowel
  - Set up HSML-Form-Website on DT Server
  - Configured API (Kafka and MySQL) to be available across Network (CSUN wifi “eduroam”)

Isaac Sim Full 5.0.0 - New Stage*

File Edit Create Window Tools Utilities Layouts Help

LIVE CACHE: NEW VERSION DETECTED

RTX - Real-Time | Perspective | Grey Studio

FPS: 68.93, Frame time: 14.51 ms
NVIDIA GeForce RTX 4070 Laptop GPU: 3.6 GiB used, 3.4 GiB available
Process Memory: 8.9 GiB used, 9.9 GiB available
1280x720

Stage | Layer | Render Settings

Search

Name (Old to New)	Type
World (defaultPrim)	Xform
CADREDemo	Xform
CADRE_Demo_Textured_material	
Measure	
flat	
CADRE_2	Xform
FR	Mesh
BR	Mesh
FL	Mesh
BL	Mesh
Chassis	Mesh
CADRE_Demo	Xform
bigRock	
_WD_Contrl	Xform
CADRE_4WD_Controllable	Xform
NewCADRE	Xform
CADRE_Textured_1	Xform
material	Scope
CADRE_Wheel_FR_1	Xform

Property

Search

Add bigRock

Prim Path /World/CADREDemo/bigRock

Instanceable

Payloads

Asset Path ./Omni_Files/Anorthosite Low Poly Extra Large Rock Set.fl

D:/School/Research/Current Research/NASA-JPL Digital Twin Interopability/Omniverse Isaac Sim/Models/Omni_Files/Anorthosite Low Poly Extra Large Rock Set.fbx

Add Payload

Semantics

This prim has no semantics

Physics

Rigid Body

Rigid Body Enabled

Kinematic Enabled

Conso... Conte...

Import

Script Editor

File Tab Edit Options Snippets

bl_omniverse_producer.py

```
if not KAFKA_AVAILABLE:
    logger.error("Kafka library not available")
    return False

try:
    logger.info("Connecting to Kafka servers: %s" % self.kafka_servers)

    self.kafka_producer = KafkaProducer(
        bootstrap_servers=self.kafka_servers,
        value_serializer=lambda v: json.dumps(v).encode('utf-8'),
        key_serializer=lambda k: k.encode('utf-8') if k else None
```

Run (Ctrl + Enter)

Ln 0, Col 0 Ins Python

Isaac Sim Full 5.0.0 - New Stage*

File Edit Create Window Tools Utilities Layouts Help

LIVE CACHE: NEW VERSION DETECTED

RTX - Real-Time | Perspective | Grey Studio

FPS: 111.99, Frame time: 8.93 ms
NVIDIA GeForce RTX 4070 Laptop GPU: 4.2 GiB used, 2.9 GiB available
Process Memory: 5.3 GiB used, 13.5 GiB available
1280x720

Stage | Layer | Render Settings

Name (Old to New) | Type

MAINDEMO
PhysicsScene

Xform
PhysicsScene
Scope
Xform
Mesh
Xform
Mesh
Xform
Xform
Mesh
Xform
Mesh
Xform
Xform
DistantLight

Script Editor

File Tab Edit Options Snippets

Python 0 | omniKafkaProducer.py

```python
# Isaac Sim Class
class OmniControls(BehaviorScript):
    def on_init(self):
        print("CONTROLS TEST INIT")
        stage = omni.usd.get_context().get_stage()
        # List of prims to track
        self.prims = [
            stage.GetPrimAtPath("/World/Omni_Cadre/CADRE_Demo/Chassis"),
        ]

    def on_play(self):
        print("CONTROLS TEST PLAY")
        for prim in self.prims:
            prim_name = prim.GetName()
            schema_id = f"schema_{prim_name}" # Use the prim's name for schema_id

            # Initialize previous state for this prim
            previous_states[prim_name] = {
                "x": None,
                "y": None,
                "z": None,
                "rx": None,
                "ry": None,
                "rz": None,
                "w": None
            }

            print(f"Tracking prim: {prim_name} with schema_id: {schema_id}")
```

Run (Ln + Enter) | Ln 0, Col 0 | Ins | Python

Console | Content

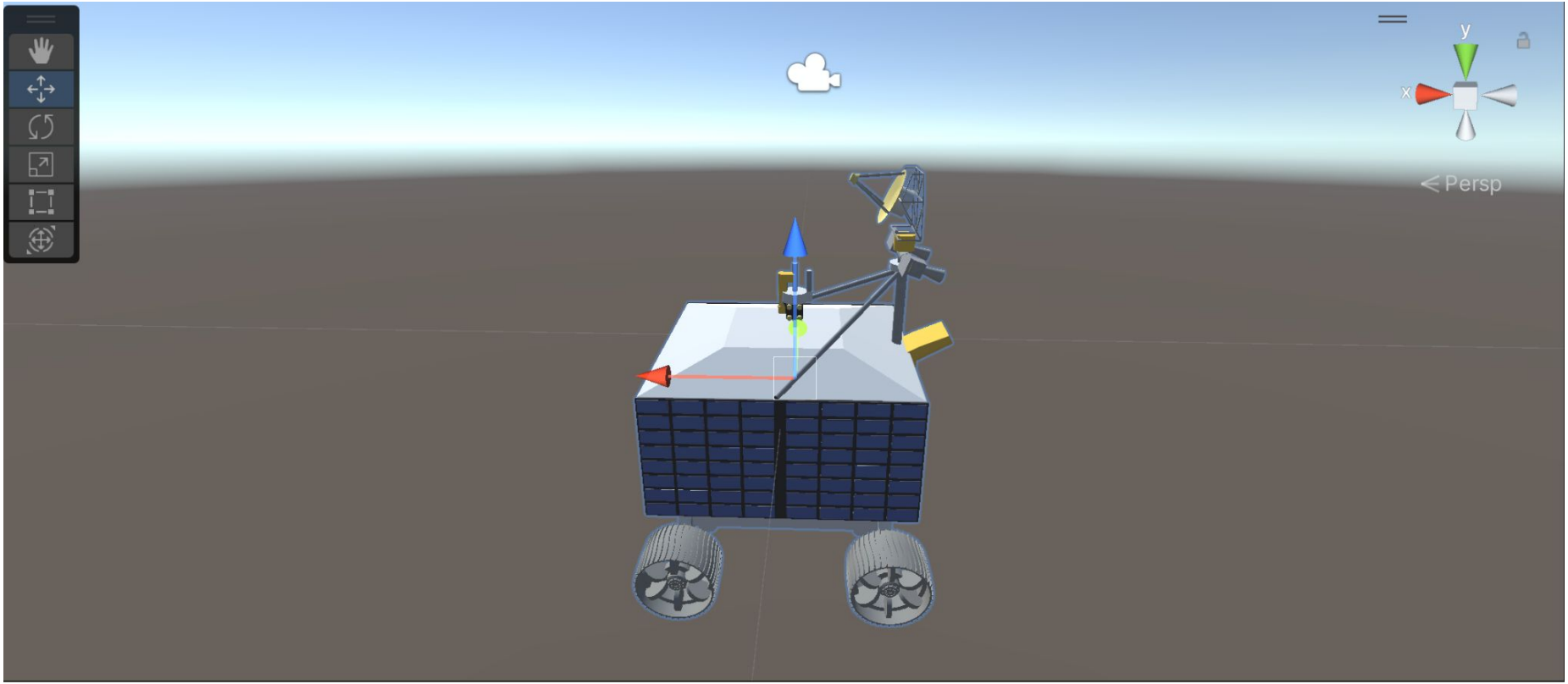
Import | D:/School/Research

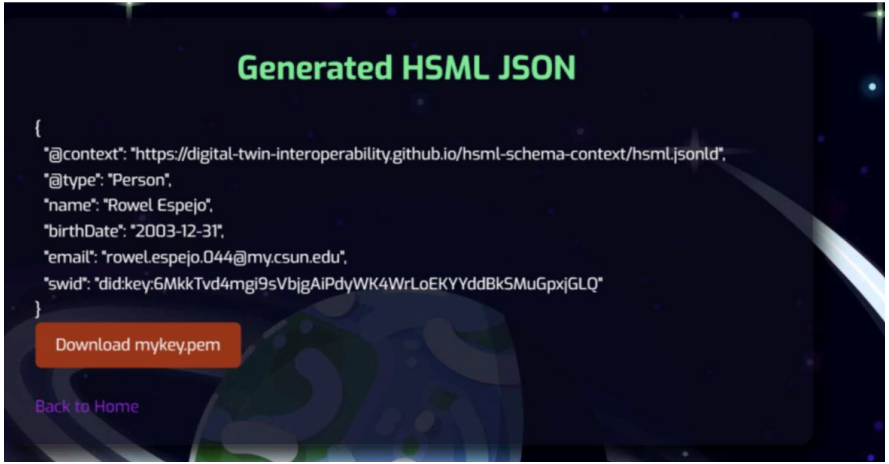
kafka_2.13
omniBL
.git
.vscode
Omniverse
DT Serv
isaac s
Models
Scripts
Setup
Setup
Papers
RAGGY
Timesheet
Unity_HSM
My Published Papers
Past Research

Location does not

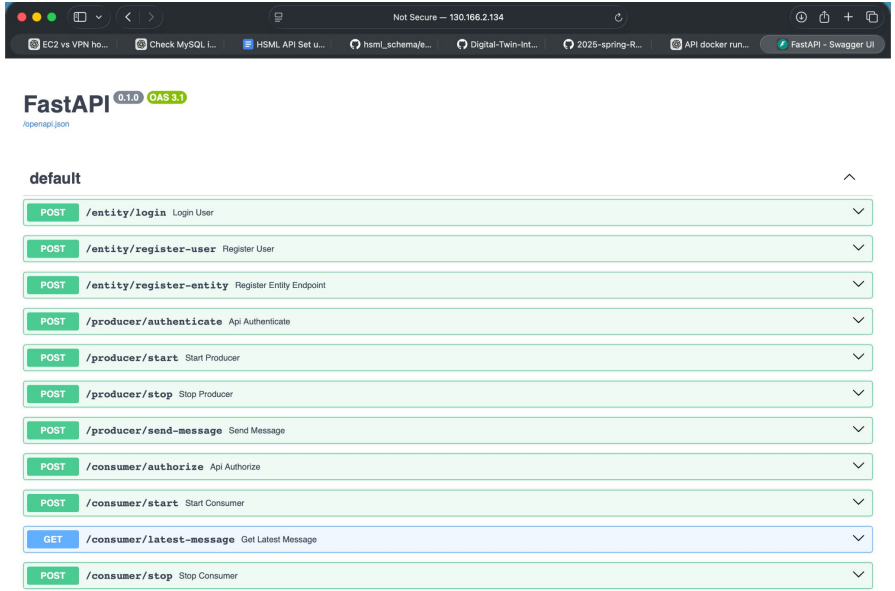
Strength Weaker than Descendants

Unity Setup





HSML-Form-Website Generating HSML JSON and Downloadable private key



Fast API Swagger UI running on local machine connected via the Network

Challenges

- Missing Unity Assets (not found in the 3D_Models_Server either):
  - Lunar_Terrain.fbx
  - Anorthosite Low Poly Extra Large Rock Set.fbx
  - Viper Substance File (1).fbx
  - Viper.fbx
  - Cadre_1.fbx
- Omniverse files:
  - Fbx files (LFS issue): Viper Substance File (1), Anorthosite file,
  - DualServer.py (missing)
- HSML-Form-Website
  - Register users not showing in did_registry

Action Plan

- Elijah
  - Connect Omniverse to Rowel’s work over network
  - Resolve missing assets ASAP
  - Continue work on BehaviorScripts and other scripts
- Martin
  - Connect unity_hsml/bl to Rowel’s work over network using csun vpn
  - Get missing assets ASAP
- Rowel
  - Continue to test the connectivity of the API over the network
  - Run Registration Test on the HSML API and HSML Web App (Refer to Testing User Manual in GitHub)